Provisional Patent Application

Grounding-Integrity Machinery for Machine-Written Knowledge: Provenance-Refusing Ingress, Quarantined Inference with Governed Promotion, Falsifier-Gated Verification, Tier-Honest Fact Serving, and Typed Non-Answers

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Field of the Invention

The present invention relates to knowledge stores and analysis engines written to and consulted by autonomous software agents, and more particularly to mechanisms that prevent machine-fabricated or machine-degraded information from masquerading as observed, verified, precise, current, or healthy: refusing, at write time and by closed vocabulary, any fact that does not declare its provenance kind; routing machine-inferred knowledge into quarantined partitions whose low trust composes through every query that touches them, promotable only by an authority-gated, audited move; refusing verification claims that do not name the observation that would have disproved them; refusing stored judgments that decay, in favor of judgments computed at read time from mandated evidence inputs; proving, at installation time and by deliberately invalid probes, that each refusal gate actually refuses; serving analysis facts under a closed precision-tier vocabulary such that an approximation can never be presented as precise and absent freshness is omitted rather than fabricated; and representing every non-answer — the unmeasurable, the unevaluated, the vacuously matched, the unreachable — as a distinct typed outcome that can never collapse into a passing one.

Background of the Invention

Large-language-model-driven agents now write facts into knowledge stores, claim to have verified their own work, and consume analysis results as grounding for further action. Each of those three activities has a failure mode in which fabricated or degraded information becomes indistinguishable from trustworthy information. The problems below are stated as technical problems in the operation of stores and analysis engines, independent of any particular agent model.

First, a fabricated fact is indistinguishable downstream from an observed one. When a deterministic parser and a language model both write into the same store, nothing in a conventional store records which writer produced which fact. A hallucinated fact — a model-invented function, a misremembered decision, a plausible but false dependency — carries exactly the shape of a true one, and every downstream consumer that trusts the store inherits the fabrication. Retrieval-augmented generation compounds the problem: the fabricated fact is retrieved as grounding for the next generation, laundering the hallucination into provenance.

Second, provenance metadata is conventionally informational, not enforced. Systems that record provenance commonly attach it as optional metadata: present when a well-behaved writer supplies it, absent otherwise, and consulted — if at all — by a reader who must remember to ask. An absent tag defaults to some reading, and either default is wrong: treating untagged as trusted fail-opens the store to every writer that forgets, and treating untagged as suspect poisons every legacy fact. Recent systems attach provenance tags to machine-inferred values and route on them — including tags distinguishing device-verified, human-reported, and model-inferred values, with routing that keeps inferred values away from decisions absent human confirmation (G. Besanson Tanasi et al., "Veritas-RPM: Provenance-Guided Multi-Agent False Positive Suppression for Remote Patient Monitoring," arXiv preprint arXiv:2604.16081, is acknowledged as prior art for such a provenance-tag taxonomy and routing discipline, in the remote-monitoring domain) — but the tag remains a routing input, not a write-time admission requirement of the store itself: nothing refuses the write that carries no tag, and nothing prevents a writer from up-tagging its own inference. What is needed is a store in which the untagged write is refused at the gate by a closed vocabulary the writer cannot extend, in which the permitted vocabulary is narrowed per fact class so that a class whose instances can only ever be declarations cannot carry an "observed" tag at all, and in which no path exists by which a producer upgrades the trust of its own output.

Third, a verification claim is itself a fabricable fact. An agent that writes "verified: the migration succeeded" into a store has written an assertion, not a verification. Nothing distinguishes a claim backed by an observation from a claim backed by nothing; a store that accepts bare verification records converts confidence into evidence. The missing discipline is falsifiability: a verification record that does not carry the observable result that would have disproved it is an assertion wearing verification's name.

Fourth, stored judgments decay. A stored fact "this execution path is live" or "this cache is current" is true at write time and silently false later — the stored verdict ages into exactly the stale fact it was meant to detect. Conventional schemas store such judgments anyway, because the alternative — computing the judgment at read time — requires the schema to have mandated, at write time, the evidence inputs the read-time computation needs.

Fifth, controls that cannot fail are read as coverage. A validation gate that is misconfigured, disabled, or never wired refuses nothing — and a store that accepts everything is indistinguishable, from the outside, from a store whose writers are all well-behaved. The inventor observed a continuous-integration gate configured to continue on error that ran for 126 seconds per build and had never once failed a build, while its existence was read as coverage. An enforcement layer whose being-on cannot be demonstrated is decoration; what is needed is a mechanism that *proves* each refusal gate refuses — by submitting deliberately invalid writes and halting if any is accepted — and that performs this proof per gate with discriminating probes, so that one gate's failure cannot make another look enforced.

Sixth, approximation is served as precision, and staleness as freshness. Code-analysis engines derive facts at different precision tiers — fast syntactic approximation, language-server-grade resolution, whole-program analysis — and serve them through one interface. A consumer that cannot see the tier treats a tree-sitter guess as an LSP-verified fact. The same engines cache and incrementally recompute, so a served fact may be current, stale, or mid-recomputation — and an interface that cannot say so serves stale facts as fresh ones. The failure has a subtle second form: an engine that *wants* to report freshness but does not yet compute it may be tempted to stamp a hardcoded "fresh" on every response, converting an honest gap into a systematic lie.

Seventh, non-answers collapse into answers. A measurement that could not be taken returns zero; a policy whose selector matches nothing reports no violations; a policy that could not run is dropped from the report; a guard evaluating an empty world finds nothing wrong; a question asked at a precision the engine cannot honor is answered at a lower precision without notice; a file that could not be read is reported as unchanged. In every case a distinct epistemic state — could not measure, never asked, could not run, nothing ingested, cannot honor, could not look — is collapsed into the healthy reading. The inventor measured a concrete instance: a blast-radius ceiling of zero denied an edit in a language the engine could parse while allowing identical edits in languages it could not, indistinguishably, because "unmeasurable" and "measured zero" shared one representation.

The inventor is not aware of a system that addresses these problems together with the properties described herein. Knowledge-graph systems that build graphs from language-model output and store provenance information for user inspection (e.g. US 2025/0131289 A1), and systems that validate language-model outputs against ground-truth data and attach citations (e.g. US 12,353,469 B1), are acknowledged as prior art for informational provenance and post-hoc output validation respectively. The present invention is directed not to informing a reader about provenance but to *enforcing* grounding integrity in the machinery itself: closed-vocabulary refusal at the write gate, algebraic quarantine at query composition, authority-gated promotion, falsifier-gated verification, installation-time gate proof, tier-honest serving, and typed non-answers.

Summary of the Invention

The invention comprises a cluster of cooperating mechanisms operating across a governed knowledge store and a structural-analysis fact server. Each mechanism is independently useful; in combination they form a system in which no path exists by which machine-written information gains trust it did not earn, loses qualification it was served with, or converts silence into health.

In one aspect, the invention provides a knowledge-ingress method in which every fact of a governed class must carry exactly one provenance kind drawn from a closed vocabulary — in one embodiment `observed` (produced deterministically from a record), `declared` (stated by a human), and `inferred` (produced by a model) — enforced by shape validation at the store's write gate such that an untagged write is refused with a message naming the missing tag; wherein the permitted vocabulary is *narrowed per class*, a class whose instances can only be declarations admitting only `declared`, and a class recording observations admitting only `observed`, so that a category error in tagging is refused as such; and wherein no interface exists by which a writer raises the provenance kind of its own prior write.

In another aspect, the invention provides quarantine of machine-inferred knowledge: facts tagged as inferred land in a reserved partition (in one embodiment a named graph) whose trust label, declared in a composable label system, is low; because label composition takes the meet across partitions touched by a query, any query result incorporating the quarantined partition carries the quarantine's low trust — the quarantine is enforced by the composition algebra, not by reader discipline. Promotion of inferred knowledge to a governed partition is a graph move requiring authority over the *target* partition, performed as an ordinary governed write —

bitemporal, attributable, auditable — and expressly not performable by the producer of the inferred content.

In another aspect, the invention provides falsifier-gated verification: a verification record is refused at the write gate unless it carries a falsifier — the observable result that would have disproved the claim — so that an assertion cannot wear verification's name; blocker records must declare their evidence strength from a closed vocabulary distinguishing a stated claim from a built demonstration, a demonstration link when present being required to reference a falsifier-gated verification record; and read-side queries report a claim whose validating verification lacks a falsifier as unvalidated, never as valid.

In another aspect, the invention provides liveness by deliberate absence: record classes whose natural "is it live / current / still true" judgment decays are defined *without* a stored judgment field — the schema instead mandates, with per-field refusal messages, the comparison inputs a read-time computation needs (in one embodiment an executing-artifact identity, a repository source, and a refresh mechanism reference), and the judgment is computed at read time by comparing mandated evidence against current state; a related read-side discipline reports dataset coverage as one of empty, none, partial, or full rather than as a bare count, so that "no producer has written yet" cannot be read as "nothing exists."

In another aspect, the invention provides installation-time gate proof: an installer that (a) writes the store configuration that turns validation-on-write on, refusing to weaken and warning rather than silently rewriting a human's contrary setting; (b) loads the ontology, shapes, and queries; and (c) *proves* each refusal gate by submitting a series of deliberately invalid writes — each probe omitting exactly one required property, so that the arms discriminate and a failure of one gate cannot make another look enforced — halting the installation with a distinct exit status if the store accepts any probe; together with a session-start banner that reports exactly one of three states — store unreachable ("could not look" — expressly not "nothing exists"), reachable but gate unproven, and active — and never blocks the session.

In another aspect, the invention provides tier-honest serving of analysis facts: every served response carries a precision tier drawn from a closed vocabulary (in one embodiment `treesitter`, `lsp`, `cpg`, `engine-state`), attached to the response envelope so that even an empty or not-found answer declares the tier at which nothing was found; the server advertises only tiers for which an extractor actually exists; a question posed at a precision the current tier cannot honor (a column-precise position under a line-precise extractor) is refused with an error rather than answered at lower precision without notice; freshness, where not computed for a surface, is omitted from that surface rather than fabricated — the response simply lacks the field, never carries a hardcoded current value — and where freshness is served (in one embodiment on a policy-verdict surface), a mid-recomputation state collapses conservatively to stale, never to fresh.

In another aspect, the invention provides typed non-answers: every distinct way an answer can fail to exist is a distinct variant of a closed result type — in one embodiment, a measurement result distinguishing measured, no-grammar-for-this-language, no-anchors, deadline-exceeded, and unreadable, wherein only the measured variant may be compared against a numeric ceiling; a change-set result distinguishing diffed, no-repository, and unresolved-reference, wherein an empty change list may be treated as clean only when every file was actually read; a policy-evaluation result in which a policy whose selector matched nothing is reported as vacuous rather than satisfied, a policy that could not run is listed as unevaluated rather than dropped, and a guard asked to evaluate a world that was never ingested refuses to evaluate rather than reporting zero violations; and an edit verifier whose verdict enumerates the violation classes it did *not* check at the current tier, so that a passing verdict cannot be over-read.

In another aspect, the invention provides fast-plane enforcement of store-governed policy at an agent's edit boundary: entities and policies are defined, versioned, and signed in the governed store; a projection engine projects the applicable policy subgraph into the fast in-memory plane of the analysis server, scoped to the agent's work, and targeted at a policy supertype so that vocabulary evolution cannot silently unbind enforcement; the edit boundary is guarded synchronously — before a proposed edit lands — by evaluating the projected policies and the structural graph against the edit, so that an edit referencing an entity that does not exist (a hallucinated reference) or violating an attached policy is rejected in the agent's loop, in real time, at a declared precision tier; the projection carries freshness, and a durable projection cache lets the guard enforce last-known policy when the governed store is unreachable while *declaring the cache's age in the verdict* rather than failing silently open or silently closed; every verdict — rejection and pass alike — is spooled and returned to the governed store as a signed fact carrying its tier and its projection freshness, the spool being retained rather than discarded when the store rejects a verdict; and an inability to project the rule plane at all is a failure surface — a status operation exits with a reserved failure code — never a silently ungoverned session.

In further aspects, the invention provides: pre-existing-condition discrimination in guarding, wherein a finding also present in the pre-action state is reported as pre-existing and attributed to no action, so that an actor is never blamed for a condition it did not cause, and blame is computed from declared effects rather than proximity; provenance-laddered authorization scope, wherein an agent's permitted scope derives from its tracked work item with scope provenance declared in a trust ladder (explicitly declared, structurally derived, observed from prior sessions) and unknown scope advises rather than blocks; and abstention as the honest resolver default, wherein work-item resolution and command parsing abstain rather than guess, because a wrong attribution manufactures false justification for derived rules.

Brief Description of the Drawings

FIG. 1 is a block diagram of a grounding-integrity system (100) according to one embodiment, showing a governed knowledge store (102) with a shape-validating write gate (104) and shapes/ontology registry (106), an autonomous writer (108) guided by a recording skill (110) through an episode write path (112), governed partitions (114) and a quarantined inference partition (116) under a composable label system (118), an authority-gated promotion move (120), a structural-analysis fact server (122) serving tier-tagged responses (124), and an installer with gate prover (126).

FIG. 2 is a diagram of the closed provenance vocabulary mechanism, showing the provenance-kind property (130), the closed value list (132), the per-class narrowing (134), and the write-time refusal with naming message (136).

FIG. 3 is a state diagram of quarantine and promotion, showing the inference partition (116) with its low trust label (140), label meet at query composition (142), and the promotion request (144) evaluated against target-partition authority (146).

FIG. 4 is a diagram of falsifier-gated verification, showing a verification record (150) with mandated falsifier (152), a blocker record (154) with closed evidence-strength vocabulary (156), and the read-side unvalidated-claims query (158).

FIG. 5 is a diagram of liveness by deliberate absence, showing a record class (160) with mandated comparison inputs (162), the read-time liveness computation (164), and coverage reporting (166) over empty/none/partial/full.

FIG. 6 is a sequence diagram of installation-time gate proof, showing the installer (126), configuration write with non-clobbering rule (170), the discriminating single-omission probe arms (172), the halt on acceptance (174), and the three-state session banner (176).

FIG. 7 is a diagram of tier-honest serving, showing extractor tiers (180), the response envelope with tier tag (124), the advertise-only-existing rule (182), the precision refusal (184), the freshness-omission rule (186), and the conservative verdict-freshness collapse (188).

FIG. 8 is a diagram of typed non-answers, showing the measurement result type (190) with its measured (192) and non-answer (194) variants, the ceiling-comparison gate (196) admitting only measured values, the vacuous and unevaluated policy outcomes (198), and the guard refusal over an empty world (200).

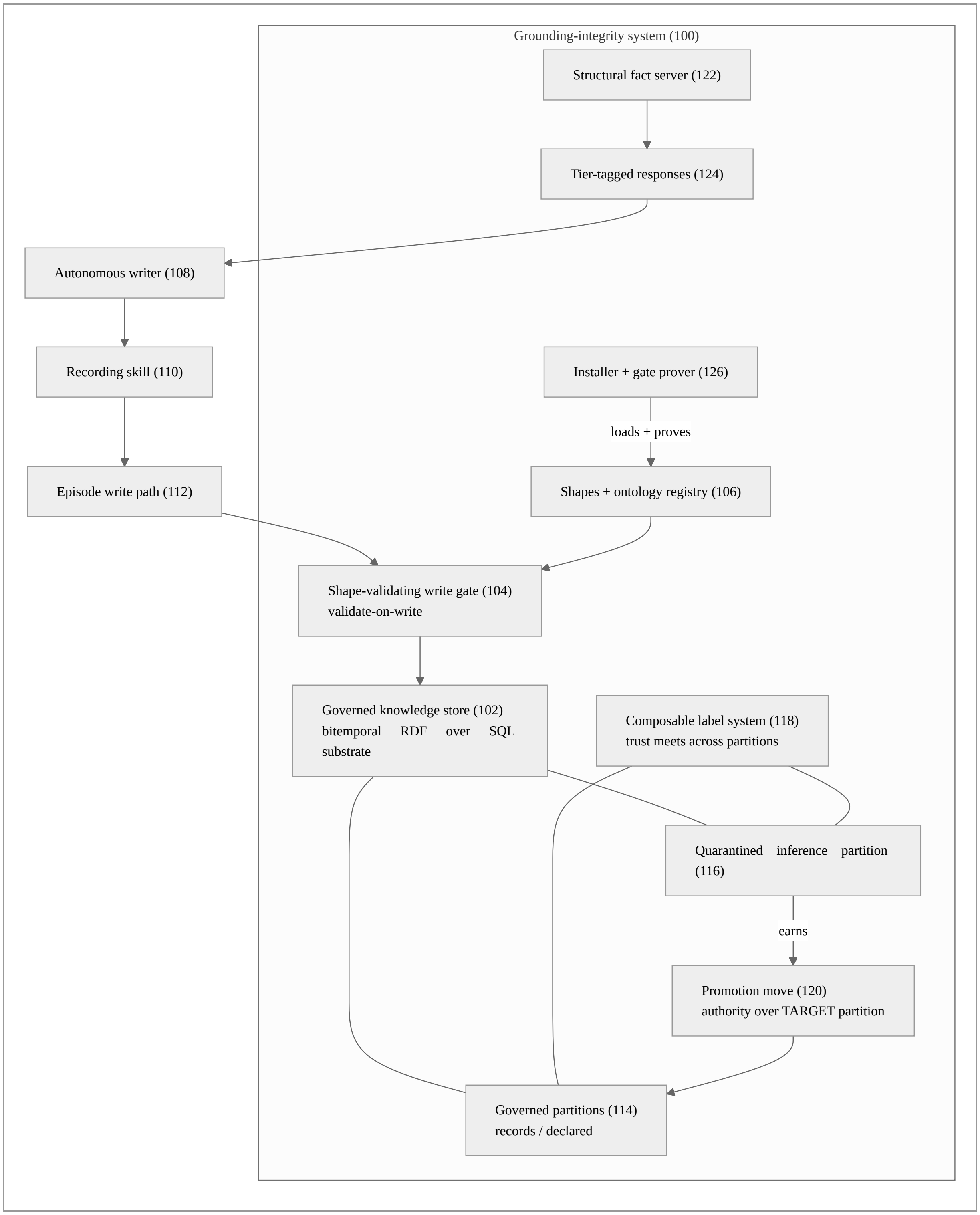
FIG. 9 is a diagram of fast-plane policy enforcement at the edit boundary, showing the governed store (102) defining entities and policies, the projection engine (210) targeting a policy supertype (212), the fast-plane projection with declared freshness (214), the durable projection cache with age declaration (216), the edit-boundary guard (218), the nonexistent-reference rejection (220), and the signed verdict return path with retained spool (222).

Detailed Description of the Invention

The following description sets forth numerous specific details to provide a thorough, enabling disclosure. It will be apparent to one skilled in the art that the invention may be practiced without these specific details, and that the specific technologies named — a particular constraint language, a particular graph data model, a particular systems programming language, particular property names — are exemplary embodiments, not limitations. Throughout, mechanisms present in the working reference implementation are so described, and mechanisms disclosed as designed combinations over that implementation are expressly identified as contemplated embodiments.

1. System overview

FIG. 1 shows a grounding-integrity system (100) according to one embodiment.



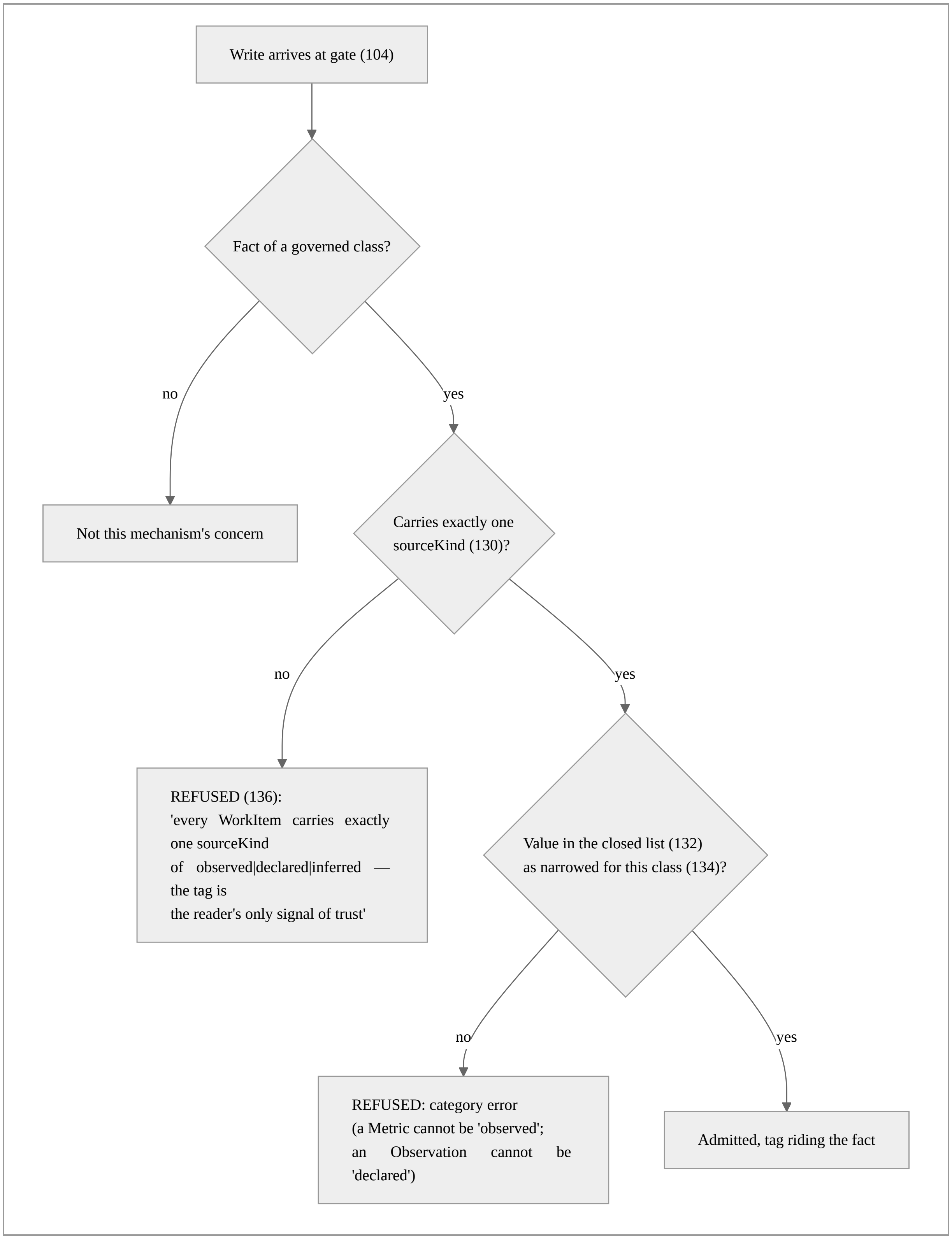
In one embodiment the governed store (102) is an embeddable bitemporal RDF knowledge-graph store supporting SHACL shape validation on write, named graphs, and composable per-graph labels; the ontology and shapes (106) are supplied by a knowledge-discipline package;

the structural-analysis server (122) is an in-memory code-analysis engine written in Rust serving parsed structure over CLI, MCP, and HTTP surfaces; and the autonomous writer (108) is a large-language-model coding agent operating under a harness that loads the recording skill (110). Facts of governed classes — work items, decisions, verification records, execution paths, blockers, and others — are written through an episode-shaped path (112) that attaches generation provenance; in the reference implementation the episode path supplies provenance-generation linkage and idempotent semantics at the store, while certain bulk ingestion paths write validated graph payloads directly, the gate (104) applying in either case. Nothing in the mechanisms below depends on these choices; alternative embodiments are identified throughout and generalized in § 10.

2. Closed, per-class-narrowed provenance vocabulary with write-time refusal (FIG. 2)

2.1 The tag is the reader's only signal of trust

Every fact class governed by the ingress discipline carries a provenance-kind property — in one embodiment `sourceKind` — whose value answers one question: *how did this fact come to exist?* The reference vocabulary is closed at three values: `observed` (produced by a deterministic parser from a record, no model in the loop), `declared` (stated by a human), and `inferred` (produced by a model). The discipline is deterministic-first: if a parser can produce the fact from a record, the parser does — because a hallucinated fact is indistinguishable downstream from a real one on exactly the facts tagged trustworthy.


Write arrives at gate (104)
Fact of a governed class?
no
yes
Not this mechanism's concern
Carries exactly one sourceKind (130)?
no
yes
REFUSED (136): 'every WorkItem carries exactly one sourceKind of observed|declared|inferred — the tag is the reader's only signal of trust'
Value in the closed list (132) as narrowed for this class (134)?
no
yes
REFUSED: category error (a Metric cannot be 'observed'; an Observation cannot be 'declared')
Admitted, tag riding the fact

2.2 Mechanism operation

In one embodiment the enforcement is a SHACL property shape per governed class: path `sourceKind`, `minCount 1`, `maxCount 1`, and an `sh:in` closed list, with a refusal message naming the class, the requirement, and the rationale. The store's write gate validates every write against the registered shapes and refuses non-conforming writes transactionally; the refusal message reaches the writer.

Three properties are deliberate:

1. **The vocabulary is closed, not extensible.** A writer cannot mint a fourth kind. `sh:in` enumerates the permitted literals; anything else — including the empty string, including a plausible synonym — is refused. An open vocabulary would let a model invent a trust-implying tag ("canonical", "confirmed") that no reader's policy anticipates.

2. **The vocabulary is narrowed per class (134).** The full three-value list applies to classes whose instances may genuinely arise any of the three ways (work items, decisions, verification records, execution paths, blockers). But a metric definition or a nonfunctional requirement is *only ever* a declaration — its shape admits only `declared` — and an observation record is only ever observed — its shape admits only `observed`. The narrowing turns a category error into a refusal: a model that writes an "observed" requirement has fabricated an observation, and the store says so at the gate rather than leaving a reader to notice.

3. **Never up-tag.** No interface exists by which a producer raises the provenance kind of its own output — there is no re-tagging write path, and the ontology documentation states the rule ("Mandatory; SHACL-refused when absent. Never up-tag."). Elevation of inferred knowledge is exclusively the promotion move of § 3, which is performed by a different authority against a different partition.

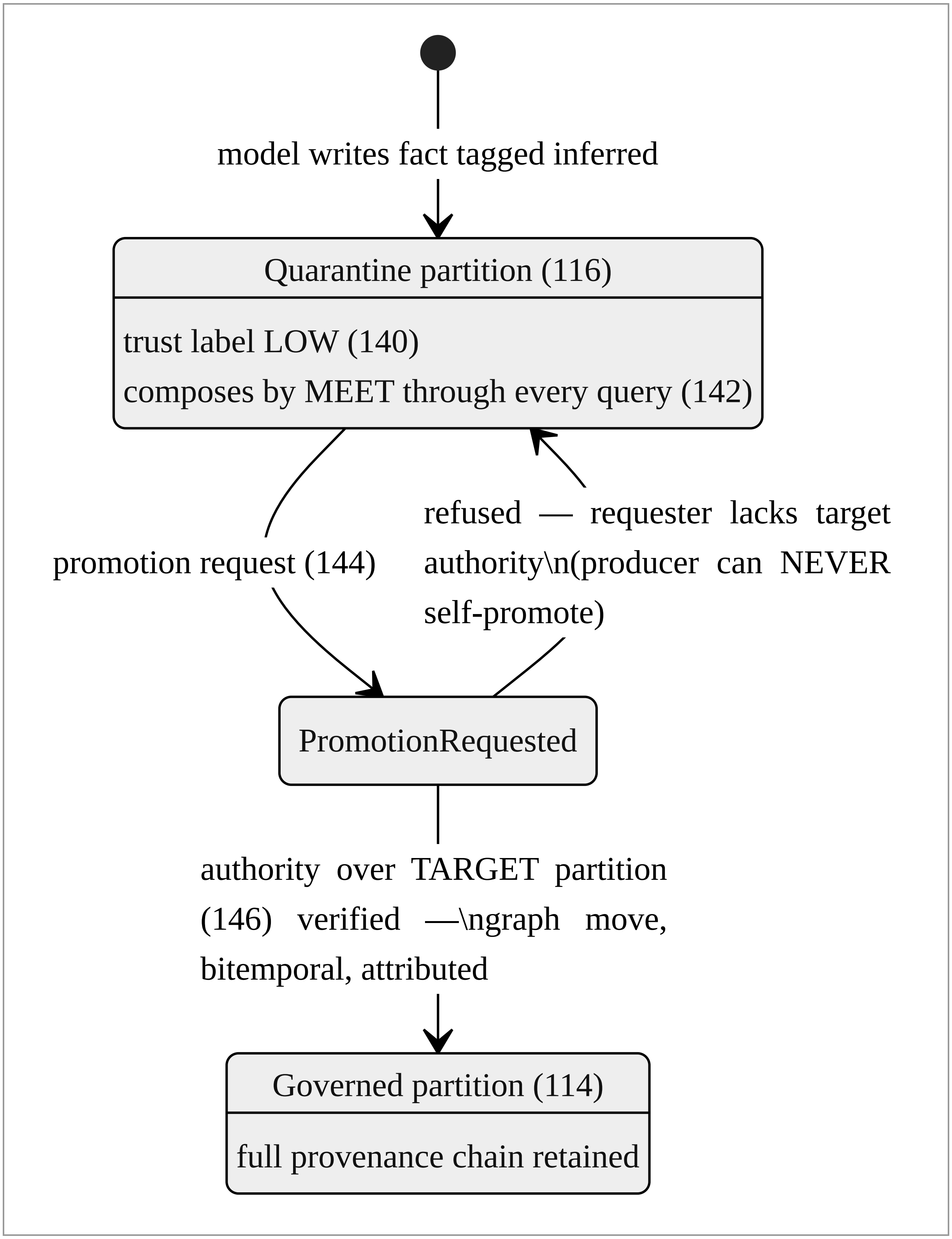
In the reference implementation the tagged classes are the crew-knowledge vocabulary (work items, decisions, verifications, execution paths, blockers, metric definitions, observations); code and document entities seeded by deterministic walkers are governed by their own shape family, their observed character being a property of the walker rather than a per-fact tag, and extending the per-fact tag to those classes is a contemplated embodiment.

In alternative embodiments the vocabulary is any closed set of provenance kinds with a defined trust ordering; the constraint language is any schema system supporting required-single-valued closed-enumeration properties with refusal (relational CHECK constraints, JSON-schema enum with additionalProperties refusal, protocol-buffer validation); and per-class narrowing is any per-type restriction of the global vocabulary to the kinds that class can truthfully bear.

3. Quarantined inference with governed promotion (FIG. 3)

3.1 Quarantine by algebra, not by discipline

Machine-inferred knowledge — session summaries, failure diagnoses, "lessons learned" — is valuable and dangerous for the same reason: it generalizes. The invention neither bans it nor trusts it: inferred facts land in a reserved partition (in one embodiment a named graph) (116) whose trust label under the store's composable label system (118) is declared low. The label system composes trust by meet — the lesser of the trust values — across every partition a query touches. The consequence is the load-bearing property: **any query whose answer draws on the quarantined partition carries the quarantine's low trust in its composed label, automatically.** A reader need not remember to check; the algebra remembers. Readers or floors that require higher trust exclude or refuse accordingly.


model writes fact tagged inferred
Quarantine partition (116)
trust label LOW (140)
composes by MEET through every query (142)
promotion request (144)
refused — requester lacks target authority\n(producer can NEVER self-promote)
PromotionRequested
authority over TARGET partition (146) verified —\ngraph move, bitemporal, attributed
Governed partition (114)
full provenance chain retained

3.2 Promotion is a governed graph move

Inferred knowledge can *earn* elevation — after human review, after corroboration by an observed fact, after a validation the deployment trusts. Promotion is deliberately not an ingress feature and not a re-tag: it is a move of the fact between partitions, executed as an ordinary governed write against the *target* partition. Because the store's authority machinery evaluates writes against the partition being written, promotion requires authority over the governed target — authority the inference producer does not hold. The move is bitemporal (the fact's history shows quarantined residence before promotion), attributable (who promoted), and auditable (the promotion is itself a fact). The skill that guides recording states the agent-facing rule: promotable later by someone with authority — never by you.

3.3 Implementation status and combination

The quarantine-and-promotion combination is disclosed as a contemplated embodiment over implemented substrate: the reference store implements named-graph partitions, composable per-graph labels with meet composition, enforcement floors, and graph-scoped write authority; the reference ingress discipline implements the `inferred` tag and the agent-facing quarantine rules; the routing of inferred-tagged writes into the reserved partition and the promotion move are specified in the ingress design and are exercised in the reference deployment as skill discipline rather than as an enforced router. An embodiment in which the write gate itself derives the target partition from the provenance kind — an inferred-tagged write to a governed partition being refused or redirected — is expressly contemplated, and is the primary embodiment for enforcement purposes.

In alternative embodiments the partitions are any data partition (tables, collections, tenants); the label system is any composable label algebra whose composition cannot raise trust; and promotion generalizes to any authority-gated move between partitions of differing declared trust, including staged multi-step promotion ladders.

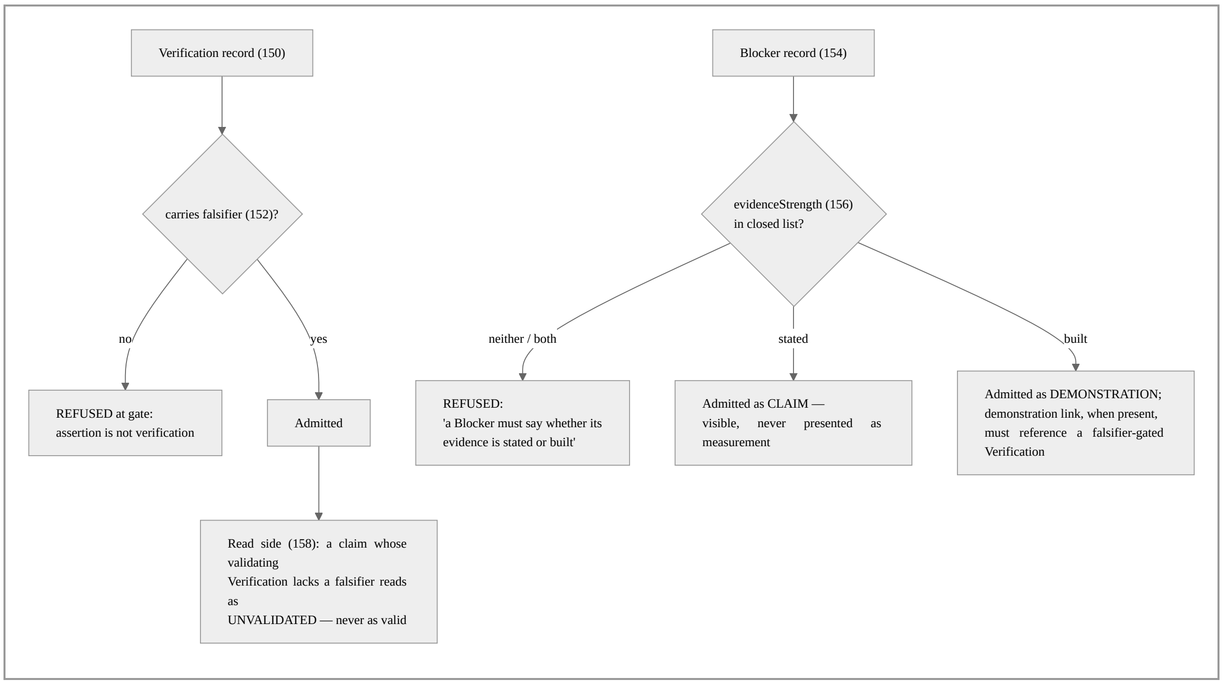
4. Falsifier-gated verification and evidence-strength-typed blockers (FIG. 4)

4.1 A verification without a falsifier is an assertion

A verification record (150) is refused at the gate unless it carries a falsifier (152): the observable result that would have proved the claim wrong. In the reference implementation this is a shape constraint (`falsifier`, `minCount 1`, `maxCount 1`, string-valued) with the refusal message:

> *a Verification without a falsifier is an assertion, not verification — record the result that would have proved it wrong*

The requirement is an ingress obligation, not a convention for a reader to remember. Its effect on machine writers is structural: an agent that "verified" something without running anything cannot name a falsifier that was capable of failing, and the record it can truthfully write is an assertion — taggable, at best, as `inferred` or `declared` — not a `Verification`.



4.2 Blockers declare their evidence strength

A blocker record (154) must declare, from a closed two-value vocabulary (156), whether its evidence is `stated` (a claim in a ticket or report) or `built` (a demonstrated failing case). The distinction lets a reader keep a stated blocker visible without presenting it as a measurement. A demonstration link (`demonstratedBy`), when present, is type-checked to reference a verification record — which, by § 4.1, carries a falsifier. In the reference implementation the demonstration link is optional (type-checked when present, not required), the producer integration that emits built demonstrations being external; an embodiment requiring a demonstration link on every `built` blocker is contemplated.

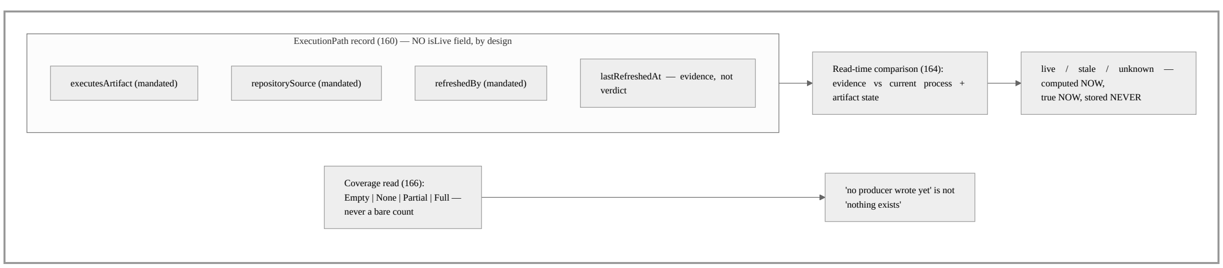
4.3 Absence reads as unvalidated

The read side completes the discipline: the reference implementation's catalogue queries report a metric claim whose validating verification is absent *or lacks a falsifier* as unvalidated — the filter requires the verification and its falsifier to exist before a claim reads as validated, so that neither a missing verification nor a gutted one can launder a claim. The general rule — absence of qualifying evidence reads as the unqualified state, never as the qualified one — is applied throughout and claimed as such.

5. Liveness by deliberate absence; coverage over counts (FIG. 5)

5.1 The schema refuses the decaying judgment

Some judgments are true only at the moment they are computed: is this execution path live, is this artifact current, is this work still in progress. A stored `isLive` field ages into exactly the stale fact it claims to detect. The invention's record classes for such subjects are defined with the judgment field *deliberately absent* — a repo-wide absence, documented in the ontology, the shapes, and the design — and with the shape instead mandating the **comparison inputs** (162) a read-time computation needs, each with its own refusal message. In the reference implementation an execution-path record must carry the executing-artifact identity (what runs), the repository source (what it was built from), and the refresh mechanism reference (what would update it); a timestamped refresh observation is evidence, never a stored current/correct verdict. Liveness (164) is then a read-time verdict: mandated evidence compared against current process and artifact state.



The refusal is two-sided: the shape refuses a record missing its comparison inputs (the read-time computation would be impossible), and the deliberate absence of the judgment field means no producer can store the decaying verdict even helpfully.

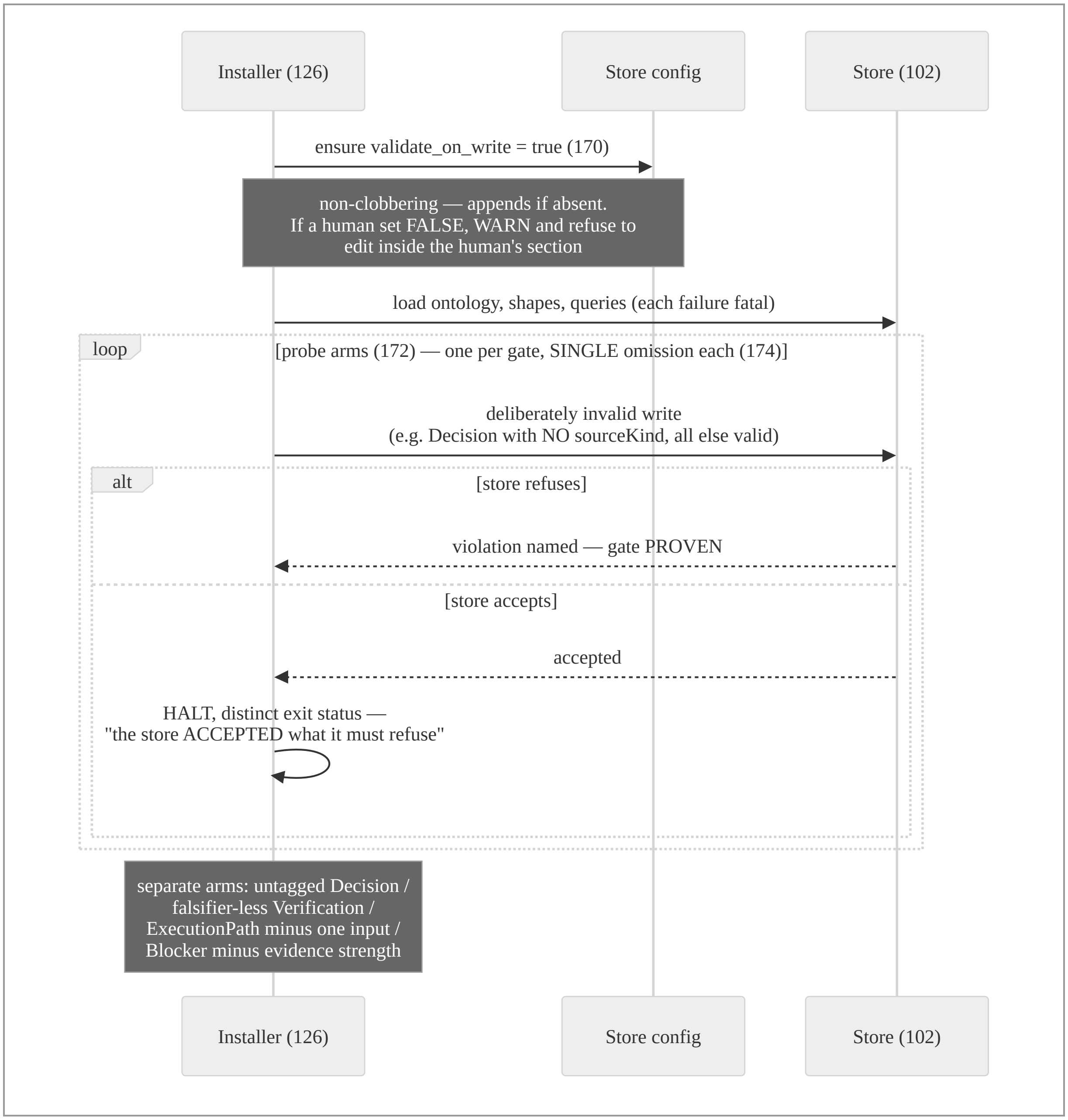
5.2 Coverage over counts

A companion read-side discipline: inventory queries report dataset coverage (166) as one of `Empty` (no dataset members at all), `None` (members present, none produced), `Partial`, or `Full`, rather than a bare count — the same coverage vocabulary the store's label system uses for label folds. A zero count collapses "no producer has written yet" and "producers wrote and found nothing" into one number; the coverage vocabulary keeps them distinct, and the reference deployment's status surfaces state the distinction ("could not look" is not "nothing exists"). In the reference implementation the coverage query executes store-side; the discipline is stated here as part of one ingress-and- reading system.

6. Installation-time gate proof (FIG. 6)

6.1 A gate that is off is a report, never a workaround

Every refusal mechanism above depends on validation-on-write being on and the shapes being loaded — a deployment property no shape can enforce about itself. The invention therefore makes gate-proof an installation-time mechanism (126):



Four properties are deliberate:

1. **The probes are discriminating: one omission per arm (174).** The untagged-write probe omits only the provenance tag; the falsifier probe carries a valid tag and omits only the falsifier; the execution-path probe omits exactly one mandated comparison input; the blocker probe omits only the evidence strength. The reference implementation records the rationale in the probe code itself: otherwise a provenance-tag failure could make a missing-falsifier shape look enforced when it never ran. Each gate is proven by a refusal only *it* can produce.

2. **Acceptance halts the install with a distinct exit status.** A store that accepts a probe is a store whose gate is off; proceeding would deploy the discipline's vocabulary without its enforcement — the "control that cannot fail" of the Background. The reference implementation prints the failure and exits with a reserved status; the installing agent's instructions are to stop, not to work around. The stated posture: a gate that is off is a report, never a workaround.

3. **The configuration write is non-clobbering toward humans (170).** The installer ensures validation-on-write is on by appending configuration when absent; when a pre-existing human-set configuration disables validation, the installer warns and refuses to edit inside the human's section — enforcement is turned on by machinery, but never by silently overriding a human's contrary record.

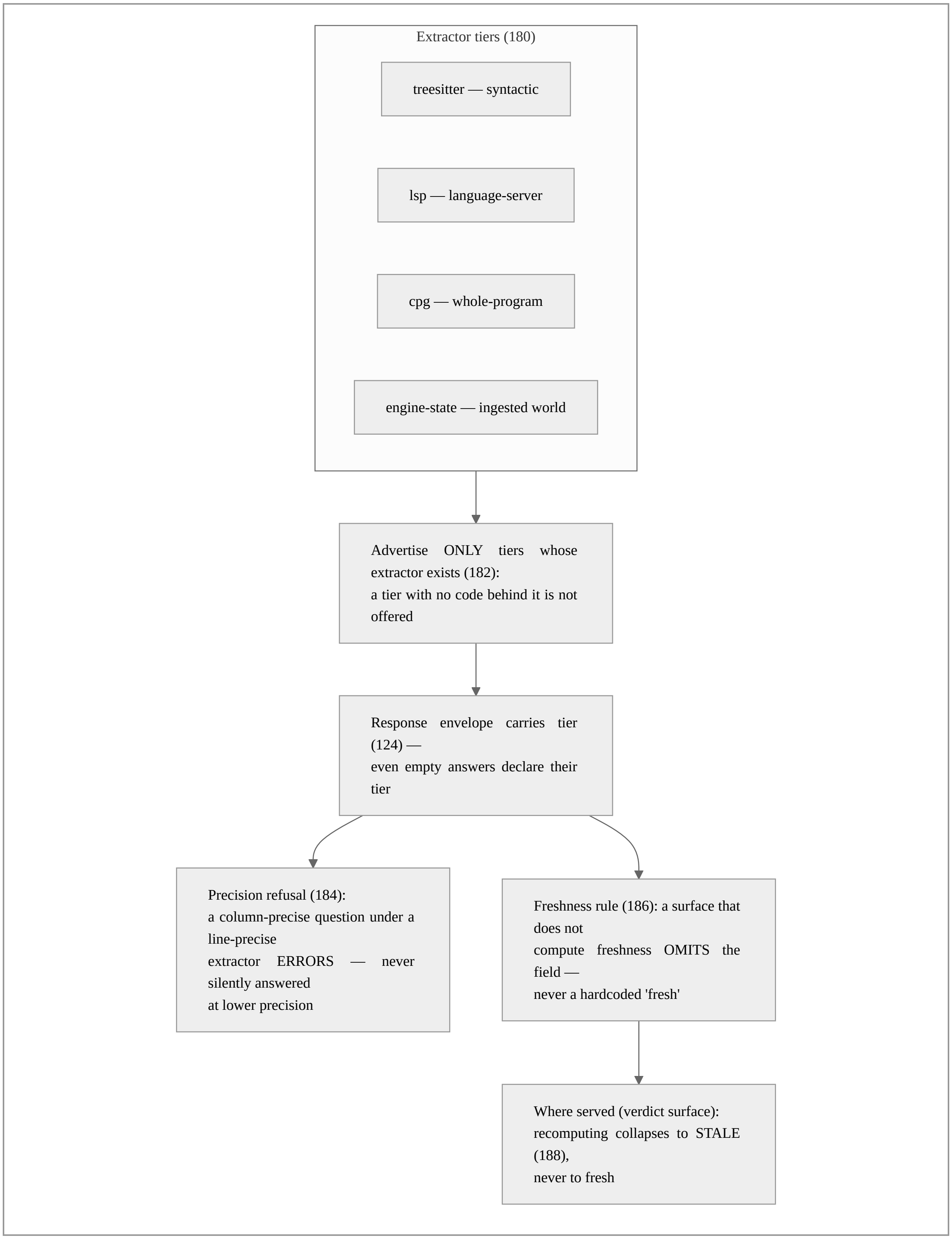
4. **The session banner is three-state and never blocks (176).** At session start a hook reports exactly one of: the store is unreachable ("could not look, not nothing exists" — a typed non-answer per § 8); the store is reachable but the discipline's shapes are not present ("the gate is not proven"); or active. All paths exit successfully — the banner informs the agent's honesty posture; it does not convert an observability gap into an outage.

In alternative embodiments the probes cover any refusal gate (constraint systems, policy engines, admission webhooks); probe arms are generated from the shapes themselves (one arm per mandatory property, expressly contemplated); the proof runs periodically rather than only at installation; and the proof's outcomes are recorded as facts in the governed store, making "when was this gate last proven?" queryable.

7. Tier-honest serving of structural facts (FIG. 7)

7.1 The closed tier vocabulary and the envelope rule

The structural fact server (122) derives facts at different precision tiers. Every served response carries a `tier` (124) from a closed vocabulary — in the reference implementation exactly four values: `treesitter` (syntactic approximation), `lsp` (language-server-grade), `cpg` (whole-program), `engine-state` (authoritatively ingested world state) — with no free-form escape. The tier rides the **response envelope**, not merely the items, so an empty or not-found answer still declares the tier at which nothing was found: "no references at the treesitter tier" and "no references" are different findings, and the envelope keeps them different.


Extractor tiers (180)
treesitter — syntactic
lsp — language-server
cpg — whole-program
engine-state — ingested world
Advertise ONLY tiers whose extractor exists (182): a tier with no code behind it is not offered
Response envelope carries tier (124) — even empty answers declare their tier
Precision refusal (184): a column-precise question under a line-precise extractor ERRORS — never silently answered at lower precision
Freshness rule (186): a surface that does not compute freshness OMITS the field — never a hardcoded 'fresh'
Where served (verdict surface): recomputing collapses to STALE (188), never to fresh

7.2 Four honesty rules

1. **Advertise only what exists (182).** The server's capability surface lists a tier only where an extractor is actually compiled in; the reference implementation removed feature flags that gated no code precisely because an advertised-but-empty tier is a masquerade at the capability level, and tests pin the advertisement to the build in both directions. A feature-gated serving surface whose engine is not built in returns not-found rather than mounting a route that cannot answer.

2. **Refuse precision you cannot honor (184).** A position query carrying file, line, *and column* under an extractor that resolves at line precision is refused with an error naming the limitation — deliberately not answered at line precision without notice, because the caller's column expresses a precision expectation the answer would silently betray. A miss is explained ("could not find what you pointed at") rather than reported as absence.

3. **Omitted, never fabricated (186).** The reference implementation tracks per-file freshness internally (fresh, stale, recomputing, with real transitions driven by a file watcher), but the graph query surfaces do not yet serve it — and the responses therefore *omit* the freshness field rather than stamping a value. The implementation history records the rule being applied to itself: an earlier revision asserted freshness was served when it was not, and the fix was to correct the claim — and separately to remove a hardcoded `fresh` from a signed verdict — rather than to fake the field. A response that cannot state freshness truthfully says nothing, and a reader must treat the absence as unknown, not as fresh.

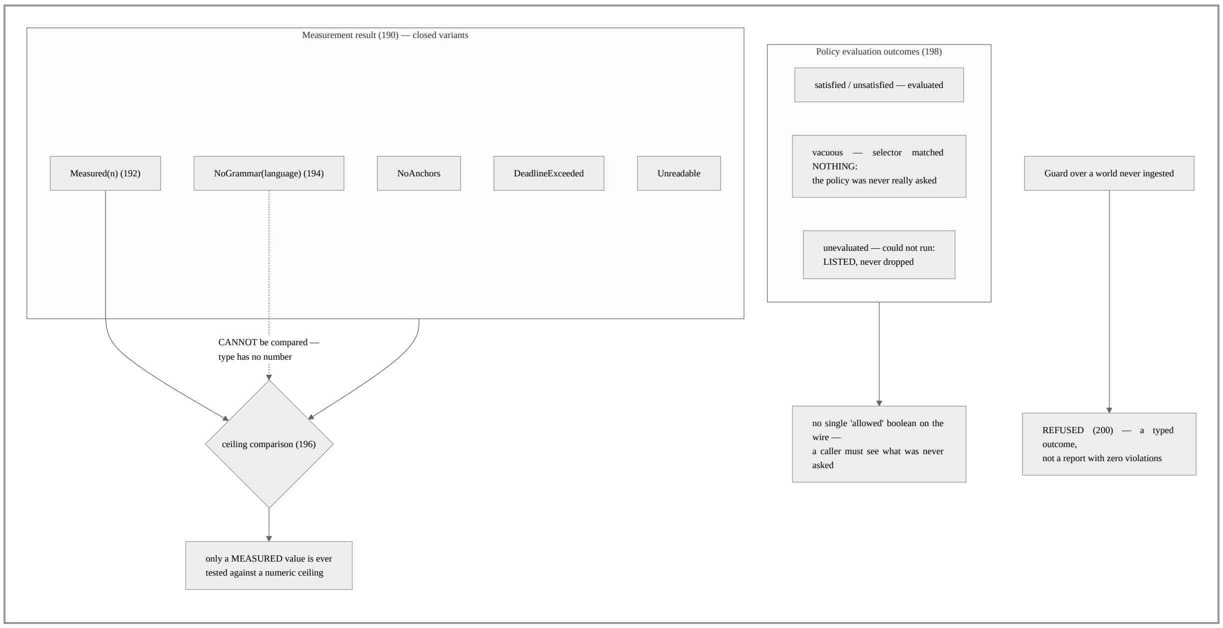
4. **Where served, collapse conservatively (188).** On the surface that does serve freshness — the policy-verdict surface, where a verdict states whether the governed rule projection it judged against is current, including its cache age — a mid-recomputation state collapses to `stale`, never to `fresh`, when the consuming schema admits only two values. The conservative reading cannot overstate. The served freshness is expressly the freshness *of the rule projection used to judge*, a different quantity from code-fact freshness; the two are kept distinct rather than conflated into one flattering number.

In alternative embodiments the tier vocabulary is any closed set of precision or derivation grades with a defined trust ordering; envelope tagging generalizes to any response schema in which the grade is mandatory at the answer level; and precision refusal generalizes to any query surface where the question's stated precision exceeds the answerable precision on any axis (position, time, identity).

8. Typed non-answers (FIG. 8)

8.1 Every way of not knowing is its own type

The system's guards and measurements return closed result types in which each distinct epistemic failure is a distinct variant — and in which the healthy variant is the *only* one that downstream numeric or boolean logic will accept.



Reference-implementation instances, each independently useful and jointly claimed as one discipline:

- **Measurement (190).** A blast-radius measurement returns one of: measured (192), no-grammar-for-this-language, no-anchors, deadline-exceeded, or unreadable (194). Only the measured variant carries a number, so only it *can* be compared against a configured ceiling (196) — the defect this repaired, a ceiling of zero denying parseable-language edits while allowing unparseable-language edits indistinguishably, is structurally unrepresentable in the typed form.
- **Change detection.** A changed-paths result distinguishes diffed / no-repository / unresolved-reference, and an empty change list may be read as "clean" only when every file was actually read — "could not look" is not "nothing changed."
- **Policy evaluation (198).** A policy whose selector matched no nodes is reported `vacuous` — a selector that rotted away from the vocabulary reads exactly like a clean report unless vacuity is a named outcome. A policy that could not run is listed

`unevaluated` — a dropped policy is indistinguishable from a satisfied one. An aggregate "allowed" boolean is deliberately not a wire field, so no caller can read one bit and skip the outcomes that were never asked.

- **Refusal over an empty world (200).** A guard asked to evaluate orders against a world nobody ingested *refuses to evaluate* — a typed refusal outcome, not a report with zero violations — because zero violations over a dead backend is a green light with nothing behind it.
- **Verification with unchecked disclosure.** The edit verifier returns, with every verdict, the list of violation classes it did *not* check at the current tier (in the reference implementation, type violations are never produced at the syntactic tier and are so listed). A passing verdict is thereby a bounded claim — "no unknown identifiers, no arity mismatches, imports resolve; types unchecked" — that cannot be over-read as full verification. The verifier's false-positive discipline is part of the mechanism: only calls the edit introduces are checked, bindings the buffer creates are excluded, and ambiguous cases are left unflagged, because a guard that cries wolf teaches its consumer to ignore refusals.
- **Pre-existing discrimination and declared-effect blame.** A guard finding also present in the pre-action state is reported pre-existing and attributed to no action; blame for a caused finding is computed from the action's *declared effects*, never inferred from proximity — an empty blame list on a caused finding is itself served as informative rather than papered over.
- **Abstention as the resolver default.** The work-item resolver and the command parser abstain rather than guess: an unknown verb, a compound command, or an unresolvable work item yields a typed abstention. The stakes are stated in the reference implementation: records are replayed to derive rules, so a wrong attribution does not mislabel one action — it manufactures false justification for a rule applied to everyone.

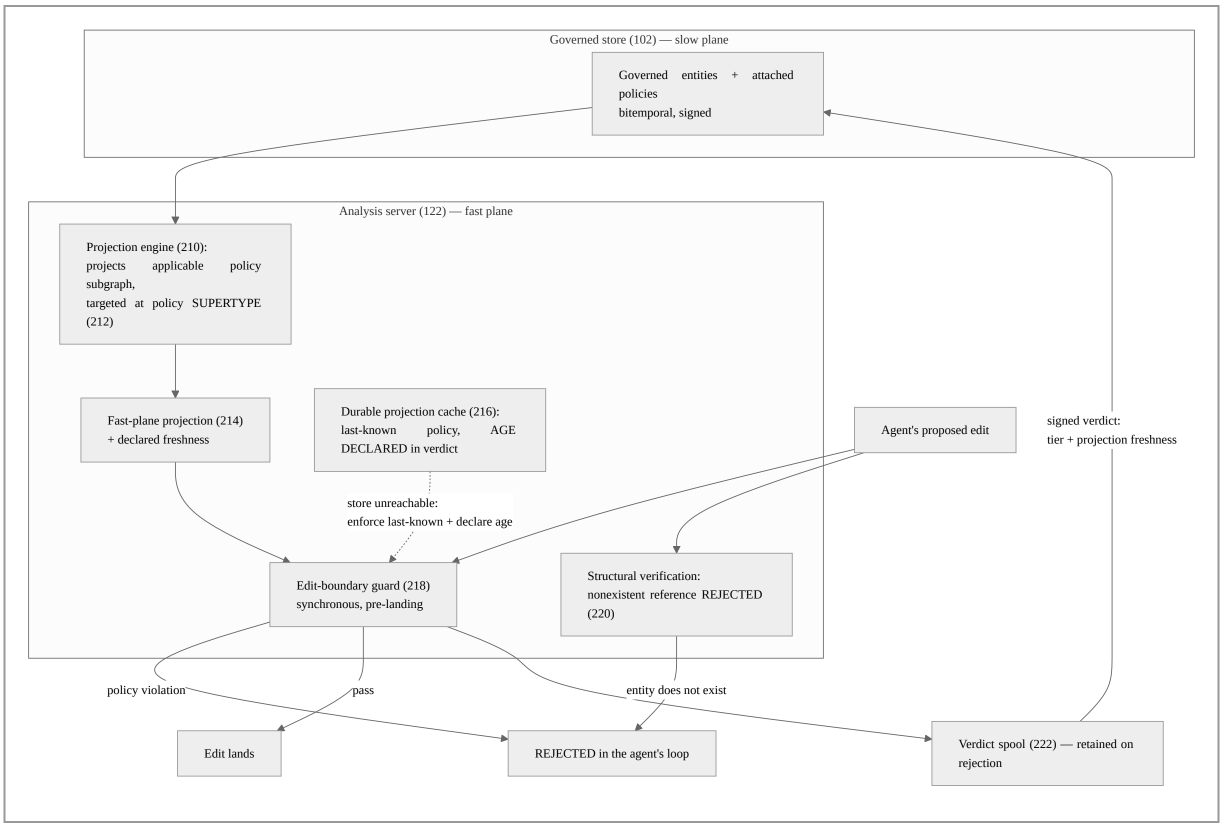
8.2 The wire and the process are honest too

The discipline extends below the result types: a client that cannot reach the resident analysis daemon receives a distinct, loud variant the calling code is forced to handle — killing one process must not silently bypass every guard. A tenant absent from the registry is distinguished from a tenant with no overlay (the latter legitimately composes the shared base). A status command exits with a reserved nonzero code when the governance rule plane cannot be projected — the rule plane is a failure surface, not a line of prose. And a promotion path that would write to a *discovered* store endpoint (rather than a configured one) refuses with a nonzero exit, so a script cannot read exploration as authorization.

9. Fast-plane enforcement of governed policy at the edit boundary (FIG. 9)

9.1 The loop this closes

An autonomous agent modifying a codebase works against entities the deployment has chosen to govern: modules, services, data surfaces — defined in the governed store with policies attached, both bitemporal and signed. The governed store is deliberately the *slow* plane: its job is authority, history, and audit, not per-keystroke latency. The agent's edit loop is the *fast* plane: a proposed edit must be judged in tens of milliseconds, synchronously, before it lands. The failure this section's machinery excludes is the gap between the planes: an agent writing code against an entity that does not exist (a hallucinated reference), or in violation of an attached policy, with the violation discovered — if ever — minutes later in review or CI, after further work has compounded on the bad edit.



9.2 Mechanism operation

1. **Projection, targeted at the supertype (212).** The projection engine (210) reads the governed store's applicable policies — including their constraint class and placement metadata — into the fast plane. The projection query targets the policy *supertype* rather

than one concrete class: the reference implementation was corrected to do so precisely because a projection bound to a concrete class silently unbinds when the vocabulary evolves — a governance layer believed on while matching nothing, the Background's fifth problem in projection form. (A vacuously empty projection is separately visible per § 8's vacuous outcome.)

2. **Scope: the projection serves the agent's work.** The fast plane is per-tenant — the analysis server composes a shared read-only structural base with the working agent's own overlay — so the subgraph the guard consults reflects *this* agent's working state, not a stale shared view; in a contemplated embodiment the projection is further narrowed by the agent's tracked work item per § 10.1, so the fast plane holds approximately the subgraph the work declares it touches.

3. **Synchronous guarding at the edit boundary (218).** A pre-action hook runs before a proposed edit lands, inside the agent harness's tool-use cycle, under a hard latency budget. It evaluates the projected policies against the edit (capability rules, structural blast-radius ceilings per § 8's typed measurements, text rules) and returns allow, deny, or notify; every decision is recorded. In the reference implementation the blocking guard is opt-in and fail-open by design at this surface — an advisory posture the deployment can harden — while the companion verification surface below is where nonexistence is rejected; an embodiment in which both run in one pre-landing gate is contemplated.

4. **Hallucinated references are rejected structurally (220).** The verification surface (§ 8, verification with unchecked disclosure) re-analyzes the proposed buffer against the base graph: a call to an identifier that exists nowhere in the composed structural graph, an arity that matches no definition, an import that resolves to nothing — each is a violation naming the symbol, returned at a declared tier with the unchecked classes enumerated. The agent's fabricated entity is refused by the structure of the code it claimed to extend, in its own loop, before the edit lands — not minutes later in review.

5. **The cache declares its age (216).** Enforcement must not depend on the slow plane's availability — but silent fallback is how guards rot. The reference implementation persists the last successful projection in a durable cache after observing that a non-persistent cache left a fraction of edits unguarded across process restarts; when the store is unreachable the guard enforces the cached policy *and states the cache's age in the verdict text,* so a verdict rendered under stale policy says so — the § 7 freshness discipline applied to the enforcement plane itself.

6. **Verdicts return to the governed store, signed (222).** Every verdict is spooled and drained to the governed store as a signed fact carrying the tier at which it was judged and the freshness of the projection it was judged under (collapsing conservatively per § 7.2). The spool is retained rather than discarded when the store rejects a verdict — evidence of a disagreement between the planes is precisely the evidence worth keeping. The governed store thus holds the audit trail of fast-plane enforcement: which edits were refused, under which policy version, at which tier, how fresh — queryable with the same temporal semantics as everything else.

7. **Ungovernable is loud.** If the rule plane cannot be projected at all, the status surface exits with a reserved failure code — the rule plane is a failure surface, not a line of prose — and the promotion path refuses to write to a store endpoint that was *discovered* rather than configured, with a nonzero exit a script cannot read as success.

In alternative embodiments the fast plane is any in-memory or edge-resident evaluation engine consuming a projection of a governed policy store (API gateways over policy databases, admission controllers over configuration stores, robotic actuator guards over safety-case stores); the edit boundary generalizes to any pre-action seam in an agent harness (tool dispatch, command execution, message send); and the structural nonexistence check generalizes to any authoritative-reference validation of a proposed action's operands against the fast plane's composed world model.

10. Supporting mechanisms and generalizations

10.1 Provenance-laddered scope; unknown scope advises

In a contemplated embodiment disclosed at the design stage, an agent's authorization scope derives from its tracked work item — the work item is the capability — and the scope's own provenance is declared in a trust ladder: `declared` (explicit edges on the item, highest), `derived` (labels, parent work, the structural graph), `observed` (what prior sessions on the item actually touched, growing with use). Unknown scope advises and never blocks, scope governs mutation rather than observation, and one scope resolution feeds three consumers (policy, trace, context). The trace substrate — work-item stamping that abstains rather than guesses, command resolution to typed action tuples, an advisory recording hook that always permits — is implemented in the reference system; the scope-resolution and enforcement model is design intent, and the pattern is claimed at the ladder-and-advisory level in the aspects.

10.2 Generalizations

- **Store substrate.** The governed store generalizes to any store with write-time schema refusal: relational engines with CHECK constraints and triggers, document stores with schema validation, any RDF store with SHACL. Named-graph partitions generalize to tables, collections, tenants, and shards.
- **Provenance vocabulary.** Three kinds generalize to any closed provenance taxonomy with a trust ordering (sensor-observed, human-attested, model-inferred, third-party-imported, …); per-class narrowing generalizes to any per-type restriction.
- **Label system.** The quarantine composes under any label algebra in which composition cannot raise trust; the meet is exemplary.
- **Falsifier.** The falsifier generalizes to any machine-checkable or human-readable statement of the disproving observation, including executable probes; blocker evidence strength generalizes to any closed claim-vs-demonstration vocabulary.
- **Gate proof.** Single-omission probes generalize to any admission-controlled system; the probe set may be derived mechanically from the schema (one probe per mandatory property).
- **Tier vocabulary.** Syntactic/semantic/whole-program/ingested generalize to any closed derivation-grade set: OCR confidence grades, sensor calibration grades, forecast model tiers.
- **Typed non-answers.** The variants generalize to any measurement or evaluation surface: the claim is the discipline — a closed result type per surface in which no non-answer variant is acceptable to logic that consumes answers — not the specific variant lists.
- **Writers.** Nothing requires the writers be machine-learning agents; the mechanisms govern any writer. The agentic setting is the motivating deployment because it maximizes the rate at which plausible falsehoods are produced and minimizes per-fact human review.

Exemplary Aspects

The following numbered aspects illustrate, in claim-like form and at several breadths, subject matter regarded as the invention. They are exemplary and non-limiting.

1. A method of admitting facts into a knowledge store, comprising: maintaining, per governed fact class, a schema requiring exactly one provenance-kind value drawn from a closed vocabulary distinguishing at least deterministically observed, human-declared, and machine-inferred origins; validating every write of a governed class against said

schema at a write gate; and refusing, with a message naming the missing or invalid tag, any write lacking exactly one permitted value, whereby no fact of a governed class exists in the store without a machine-checked provenance kind.

2. The method of aspect 1, wherein the closed vocabulary is narrowed per fact class such that at least one class admits only a single provenance kind, a write tagging an instance of that class with any other kind being refused as a category error.

3. The method of aspect 1, wherein no interface of the store permits a writer to raise the provenance kind of a previously written fact, elevation of machine-inferred content being exclusively a partition-move operation under aspect 5.

4. The method of aspect 1, further comprising routing facts bearing the machine-inferred kind into a reserved partition whose declared trust label, under a composable label system in which composition cannot raise trust, is lower than that of governed partitions, whereby any query result drawing on the reserved partition carries the reduced trust in its composed label without reader intervention.

5. The method of aspect 4, further comprising promoting a fact from the reserved partition to a governed partition only by a move operation authorized against the target partition, recorded with attribution and temporal history, the producer of the fact holding no authority to perform the move.

6. A method of recording verification in a data store, comprising: refusing, at a write gate and by schema, any verification record that does not carry a falsifier field stating the observable result that would have disproved the verified claim; and reporting, on the read side, any claim whose associated verification record is absent or lacks a falsifier as unvalidated, absence of qualifying evidence never being reported as validation.

7. The method of aspect 6, further comprising requiring each blocker record to declare exactly one evidence strength from a closed vocabulary distinguishing a stated claim from a built demonstration, and type-checking any demonstration link to reference a falsifier-bearing verification record.

8. A method of representing decaying judgments in a data store, comprising: defining a record class for a subject whose liveness-like judgment changes after write time without a stored judgment field; requiring instead, by schema with per-field refusal, the comparison inputs sufficient to compute the judgment at read time, including at least an artifact identity and a source reference; and computing the judgment at read time by

comparing the mandated inputs against current system state, whereby no stored verdict can age into the stale fact it purports to detect.

9. The method of aspect 8, further comprising reporting inventory coverage as one of empty, none, partial, and full rather than as a bare count, a dataset with no producing members being reported distinctly from a dataset whose members produced nothing.

10. A method of installing an enforcement discipline onto a data store, comprising: ensuring a validation-on-write configuration is enabled, appending configuration where absent and, where a human-authored configuration disables validation, warning and declining to modify the human-authored setting; loading schemas into the store; submitting a plurality of deliberately invalid probe writes, each probe omitting exactly one required property of one schema such that each gate is proven by a refusal only it can produce; and halting the installation with a distinct failure status upon the store accepting any probe.

11. The method of aspect 10, further comprising reporting, at the start of an agent session, exactly one of at least three states — the store unreachable, the store reachable with the discipline's schemas absent, and the discipline active — wherein the unreachable state is reported as inability to observe rather than as absence of enforcement, and no state blocks the session.

12. The method of aspect 10, wherein the probe set is derived mechanically from the loaded schemas, one probe per mandatory property.

13. A method of serving analysis facts, comprising: deriving facts at a plurality of precision tiers; attaching to every served response, including empty and not-found responses, exactly one tier value from a closed vocabulary; and advertising, on the server's capability surface, only tiers for which a deriving extractor is present in the running build.

14. The method of aspect 13, further comprising refusing, with an error identifying the limitation, any query whose stated precision on any axis exceeds the precision the present tier can honor, the query never being answered at a lower precision than stated without notice.

15. The method of aspect 13, further comprising omitting a freshness field from any served surface for which freshness is not computed, a fabricated or hardcoded freshness value never being served, and, on a surface where freshness is served with a narrower

consuming vocabulary than the internal state, collapsing an intermediate recomputing state to stale and never to fresh.

16. The method of aspect 15, wherein a freshness value served with a policy verdict denotes the currency of the rule projection used to judge, stated together with the projection's age, and is not conflated with the freshness of the underlying analyzed facts.

17. A method of returning measurements in a guarded system, comprising: returning, from a measurement operation, one variant of a closed result type distinguishing a measured value from each of a plurality of non-answer conditions including at least unparseable-input and deadline-exceeded; and comparing against any configured numeric threshold only the measured variant, the non-answer variants carrying no number and thereby being structurally incapable of satisfying or violating the threshold.

18. A method of reporting policy evaluation, comprising: reporting a policy whose selector matched no nodes as vacuous rather than satisfied; listing a policy that could not be evaluated as unevaluated rather than omitting it; refusing to evaluate, with a typed refusal outcome, a request against a world state that was never ingested; and omitting from the wire format any aggregate boolean that would permit a consumer to accept a result without observing which policies were never asked.

19. A method of verifying a proposed edit against an analysis graph, comprising: checking the edit at a stated precision tier for at least references to identifiers that do not exist and mismatched arity; returning with the verdict an enumeration of violation classes not checked at that tier; and excluding from violation any call site the edit did not introduce and any name the edited buffer itself binds, whereby a passing verdict is a bounded claim that cannot be over-read and refusals are reserved for violations the edit itself would introduce.

20. A method of attributing guard findings, comprising: re-evaluating each finding against the pre-action state; reporting a finding also present pre-action as pre-existing and attributing it to no action; and computing blame for caused findings from effects declared by the actions rather than from proximity, an empty blame attribution being served as informative rather than suppressed.

21. A method of scoping an autonomous agent's authority, comprising: deriving the agent's permitted scope from a tracked work item; labelling each scope element with a scope provenance from a closed ladder distinguishing at least explicitly declared, structurally

derived, and observed-from-prior-sessions origins; and, where scope is unknown, advising rather than blocking, scope being applied to mutating actions and not to observation.

22. The method of aspect 21, wherein a work-item resolver and a command parser abstain, returning a typed abstention rather than a guess, when resolution is ambiguous, records so produced being used to derive policy such that a false attribution would manufacture justification for a derived rule.

23. A method of enforcing store-governed policy at an autonomous agent's action boundary, comprising: maintaining entities and policies in a governed store; projecting an applicable policy subgraph into an in-memory fast plane of an analysis server, the projection targeting a policy supertype such that evolution of concrete policy classes cannot silently unbind the projection; evaluating, synchronously and before a proposed action lands, the projected policies and a structural graph against the proposed action; rejecting, in the agent's loop and at a declared precision tier, an action that violates a projected policy or that references an entity absent from the structural graph; and recording every evaluation outcome.

24. The method of aspect 23, further comprising persisting the most recent successful projection in a durable cache, and, when the governed store is unreachable, enforcing the cached projection while declaring the cache's age within the rendered verdict, the guard neither silently passing nor silently enforcing stale policy unannounced; and treating inability to project the policy plane as a reserved failure status of a status operation rather than as an ungoverned session.

25. The method of aspect 23, further comprising returning each verdict to the governed store as a signed fact carrying the tier at which it was judged and the freshness of the projection under which it was judged, and retaining, rather than discarding, spooled verdicts that the governed store rejects.

26. A system comprising a processor and storage configured to perform the methods of aspects 1, 4, 6, 8, 10, 13, 17, 18, and 23 in combination, wherein a governed knowledge store enforces the provenance, verification, and liveness schemas of aspects 1, 6, and 8 at a validating write gate proven by the method of aspect 10, machine-inferred facts are quarantined and promoted per aspects 4 and 5, an analysis fact server consulted by the same agents serves tier-tagged, freshness-honest, typed-non-answer responses per aspects 13, 17, and 18, and the same server enforces the store's policies at the agents'

edit boundary per aspects 23 through 25, whereby no path exists in the combined system by which machine-written information is admitted untagged, presented above its precision, aged into falsehood, read as healthy through silence, or landed as an edit against an entity that does not exist or a policy that forbids it.

The foregoing is a provisional disclosure. No formal claims are presented; the Exemplary Aspects above illustrate the subject matter regarded as the invention. Provenance-tag taxonomies with routing (Veritas-RPM, arXiv:2604.16081), informational provenance in LLM-constructed knowledge graphs (US 2025/0131289 A1), and post-hoc validation of model outputs against ground truth (US 12,353,469 B1) are acknowledged as prior art for their respective concepts; the mechanisms disclosed and aspected above are enforcement mechanisms — write-gate refusal, algebraic quarantine, authority-gated promotion, installation-time gate proof, tier-honest envelopes, and typed non-answers — distinct from those concepts.